1 Introduction

This study estimates employment from Angul's assessed 10,805 MW of solar potential using published coefficients, extended with skill, gender and rooftop data. The headline result is about 8,200 one-off job-years and 6,600 FTE of O&M, with floating solar providing 70% of O&M. This is only about 4% of the 168,721 coal-dependent workers, so solar can complement but not replace coal livelihoods without training and local hiring.

2 Data sources

Only published secondary data are used. **Solar potential** (iFOREST [1], Table 8, p. 107): 7,505 MW ground-mounted (934 MW reclaimed overburden, 4,646 MW excavated mine land, 1,925 MW wasteland) and 3,300 MW floating; 26 MW rooftop is kept as an extension. This is assessed potential, not a project pipeline. **FTE coefficients** (CEEW-NRDC [2], Figs 4 to 6), in FTE job-years per MW by phase (business development, design, construction, O&M): ground 0.18, 0.11, 0.45, 0.26 (total 1.00); floating 0.05, 0.18, 0.58, 1.40 (2.21); rooftop 17.30, 4.12, 13.93, 9.33 (44.68). **Skill shares** by phase (Figs 7 to 9) and **women's share** of workforce (Fig. 25: ground 11%, floating 12%, rooftop 15%) come from the same report. **Coal workforce** (iFOREST, Ch. 4, Table 8): 168,721 direct workers, 116,493 of them informal. Coefficients are national company-survey averages (floating rests on 3 companies, ground on 19), so results are scenarios, not forecasts. All outputs are FTE job-years (one-off) or FTE per operating year (O&M), not permanent jobs.

3 Method and code

For technology k, deployment share $s \in \{0.25, \ldots, 1\}$ and phase p: $J_{k,p} = sP_k\beta_{k,p}$, $J_{\text{total}} = \sum_k \sum_p J_{k,p}$. The script `angul_solar_employment.py` (pandas, numpy, matplotlib):

1. stores all inputs in dictionaries with source comments;
2. `phase_jobs`, `totals` compute $J_{k,p}$ and split one-off (first three phases) from recurring O&M;
3. `annual_profile` adds sP/T MW each year for $T = 10, 20, 30$ years, spreads one-off effort over those years and gives each yearly batch 25 years of O&M (our assumption);
4. `sensitivity` changes each coefficient by ±30%; `skill_breakdown`, `women_estimate`, `rooftop_extension` and `coal_comparison` are the extensions;
5. `validate` asserts that the output equals the earlier project numbers (14,798; 8,227; 6,571), then the tables (CSV) and four figures are saved.

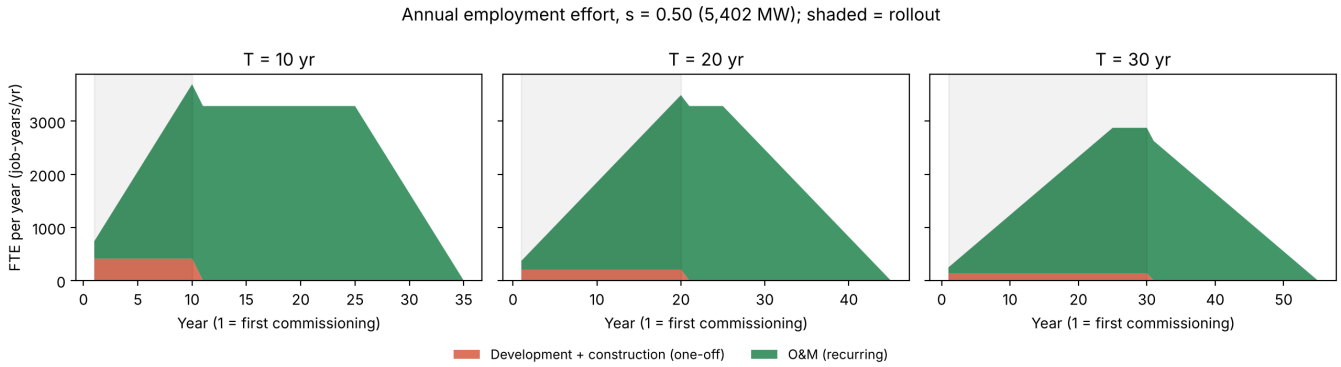


Figure 1: Yearly employment effort at $s = 0.5$ for three rollout periods (shaded = rollout). Red: one-off development and construction; green: recurring O&M.

4 Results and analysis

$s = 1$	Ground	Floating	Total
Business dev.	1,351	165	1,516
Design	826	594	1,420
Construction	3,377	1,914	5,291
One-off subtotal	5,554	2,673	8,227
O&M (FTE/yr)	1,951	4,620	6,571
J_{total}	7,505	7,293	14,798

Table 1: $J_{k,p}$ at full assessed potential.

s	MW	One-off	O&M	Peak T=30
0.25	2,701	2,057	1,643	1,369
0.50	5,403	4,113	3,286	2,738
0.75	8,104	6,170	4,928	4,107
1.00	10,805	8,227	6,571	5,476

Table 2: Scenario totals (job-years; FTE for O&M).

- **Scale.** Full potential gives 8,227 one-off job-years and 6,571 FTE of O&M. Floating solar is 30.5% of MW but 70.3% of O&M, because of its O&M coefficient (1.40 against 0.26 FTE/MW). Treating everything as ground-mounted

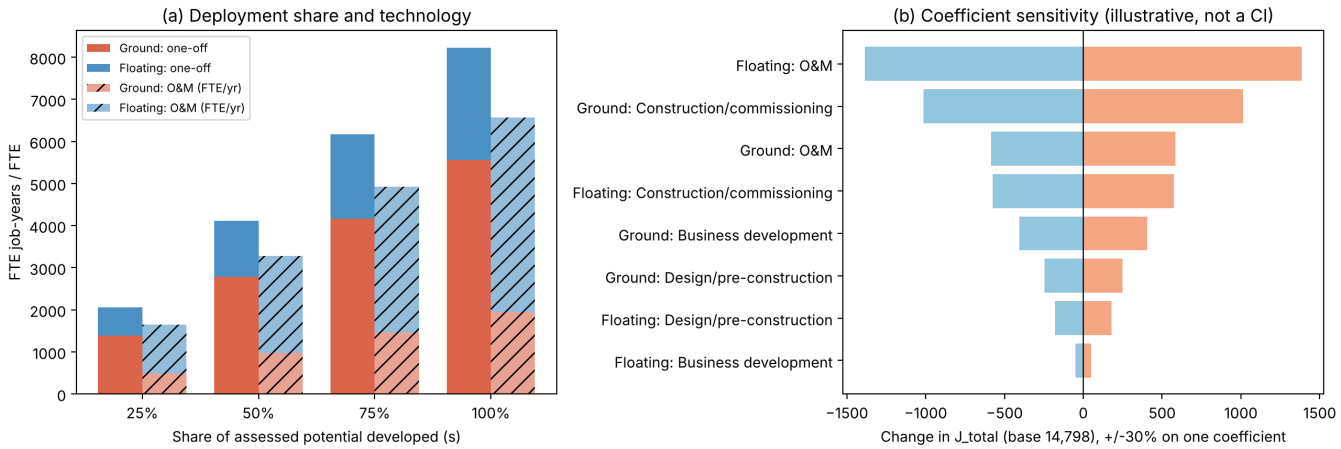


Figure 2: (a) One-off job-years and O&M FTE by deployment share and technology. (b) Change in J_{total} for ±30% on one coefficient (illustrative, not a confidence interval).

would understate J_{total} by 27%.

- **Timing.** Totals scale linearly with s; T changes only timing. At $s = 0.5$ one-off effort is 411, 206 and 137 FTE/yr for $T = 10, 20, 30$. For $T = 30$ the O&M peak falls (2,738 against 3,286) as early batches retire. O&M is about 95% of lifetime job-years (82,141 of 86,255).
- **Sensitivity.** Floating O&M is the largest driver (±1,386, 9.4%), then ground construction (±1,013, 6.8%). The biggest driver rests on the thinnest sample.
- **Skills.** O&M is 24% highly, 31% semi and 45% low skilled; one-off work is 24%, 39%, 37%. Construction is the most low-skilled phase (59% ground, 55% floating), so civil work and transport may overlap with coal-sector skills.
- **Gender.** Women hold 11 to 12% of workforce roles in these sectors. If this applied to field work, about 770 O&M FTE and 930 one-off job-years would go to women. This is an upper bound since the share covers all roles.
- **Rooftop.** 26 MW adds about 919 one-off job-years and 243 O&M FTE, because the coefficient is 44.68 FTE/MW.
- **Coal comparison.** Full-build O&M (6,571 FTE) is about 3.9% of coal-dependent workers and 5.6% of informal ones; its low-skilled part (2,955 FTE) is 2.5% of informal workers. FTEs are not headcounts, so this shows scale only.
- **Feasibility.** About 52% of assessed MW is on mine land available only after mine closure. The only named floating project (300 MW, Rengali) is about 9% of the floating potential. Skill transfer of coal workers is possible, not shown by the data.

Conclusion. Solar can complement but is unlikely to replace coal-dependent livelihoods in Angul. It needs training and local-hiring measures.

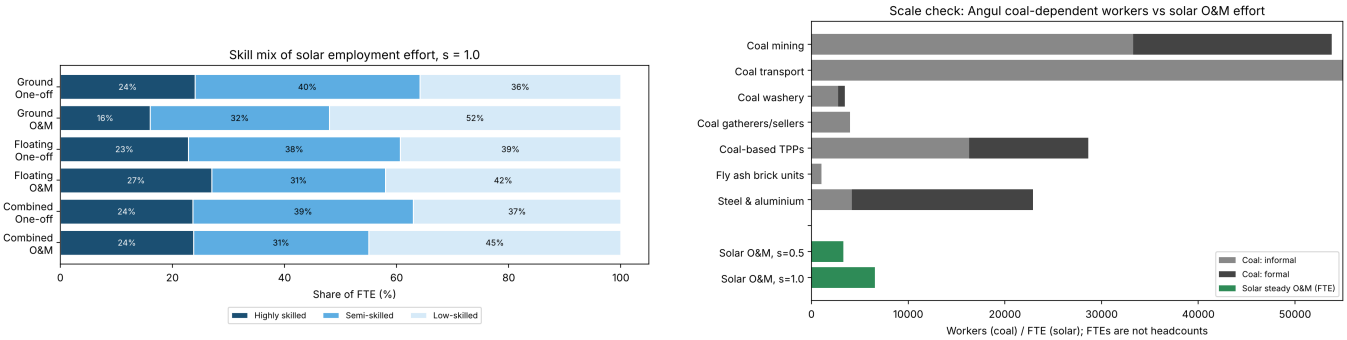


Figure 3: Left: skill mix of solar employment by technology and work type ($s = 1$). Right: coal-dependent workers by category against steady solar O&M effort (workers against FTE, scale only).

References

[1] iFOREST (2022). *Angul: Planning a Just Energy Transition and a New Green Economy*. https://iforest.global/report/angul-planning-a-just-energy-transition-and-a-new-green-economy/

[2] Almeida, N. et al. (2026). *Driving Energy Transition: Workforce, Skills, and Gender in India's Renewable Energy Sector*. CEEW-NRDC. https://www.nrdcindia.org/pdf/Driving-Energy-Transition_2026_Web-Optimised.pdf